



Machine Learning Operations

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Exam Example

Name of the student:

Number:

Duration of the exam: **60 minutes**

In each question there exist the information of how much each question worth in a scale of **0 to 20 pts**. A multiple-choice wrong answer corresponds to a certain penalty. In each question you will see (score/ -penalty). Write the capital letter correspondent to the right answer.

- 1- One of the key features of MLOps is having a specialized structure in each relevant component of our framework. A mature organization should have well defined roles for each stage of the MLOps framework. Which role is better suited for model deployment to productions? (1 pts/ -0.25pts)

- A) Data engineers DevOps
- B) Data Scientist.
- C) Data engineers.
- D) Subject matter expert.

Choose the right option: ____ A

- 2- What solution from DevOps is one of the key features of MLOps general framework? (1 pts/ -0.25pts)

- A) Implementation of software.
- B) Continuous Integration and Continuous delivery.
- C) Defined roles in the framework.
- D) Model experimentations.

Choose the right option: ____ B

- 3- Model Governance is a new challenge in our present. What will have a great impact on the integration between Machine Learning governance and MLOps? (1 pts/ -0.25pts)

- A) Size of the data.
- B) Regulations and number of ML models.
- C) Type of software used to create the models.
- D) Feature selection method.

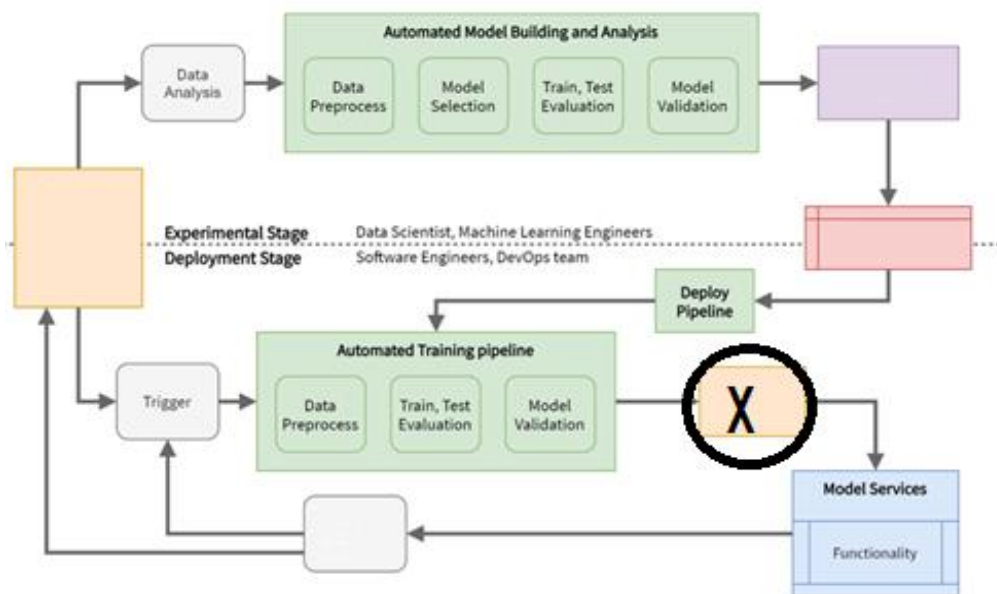
Choose the right option: ___ B

- 4- While having a model in production, we still keep the development of new models and we can put the new ones as challengers for the champion model, inferring on the same events as the champions model, the shadow testing. When it is not possible perform shadow testing? (1 pts/ -0.25pts)

- A) The ground truth cannot be evaluated for both models.
- B) The two models are different: one can be a tree model and the other is a neural network.
- C) The models have different features as inputs.

Choose the right option: ___ A

- 5- This figure represents a procedure of a MLOps pipeline. What is the option that represents the process X? (1 pts/ -0.25pts)



- A) Data tests.
- B) Model Registry.
- C) Git operations.
- D) Pytest run.

Choose the right option: ___ B

- 6- For which process is the **requirements.txt** fundamental? (1 pts/ -0.25pts)

- A) Creating versioned models.
- B) Creating virtual environments.

C) To define unitary data tests.

Choose the right option: ____ B

7- What is the main difference between covariate shift and label drift? (1 pts/ -0.25pts)

A) Covariate shift is a consequence of label drift.

B) Covariate shift is related to a change in the distribution of the input variables of the model. The label drift corresponds to a change in the distribution of the target variable of the model.

C) Label drift is related to a change in the distribution of the input variables of the model. The covariate shift corresponds to a change in the distribution of the target variable of the model.

Choose the right option: ____ B

8- What kind of drift, COVID-19 generated? (1 pts/ -0.25pts)

A) Sudden drift.

B) Gradual drift.

C) Incremental drift.

D) Recurring concepts.

Choose the right option: ____ A

9- What is the main difference between the Kolmogorov-Smirnov test and the Jensen-Shannon test? (1 pts/ -0.25pts)

A) Kolmogorov-Smirnov is used only to detect cases of concept drift, while Jensen-Shannon can be used for all tests.

B) Kolmogorov-Smirnov is based in measurement of the relative entropy between two distribution and Jensen-Shannon is a nonparametric statistical test where null hypothesis that two samples come from the same distribution is evaluated.

C) Jensen-Shannon is based in measurement of the relative entropy between two distribution and Kolmogorov-Smirnov is a nonparametric statistical test where null hypothesis that two samples come from the same distribution is evaluated.

Choose the right option: ____ C

10- In a fraud detection problem, after the model is in production, we need to evaluate its performance. Since there is a delay on the fraud classification, our metrics cannot be the most accurate ones. How we can mitigate this risk in the short term, and check the true performance of our model? (1 pts/ -0.25pts)

A) Create a second model and check the intersection of both blocked transactions.

- B) Wait more time until the fraud starts to be classified.
- C) Create a group control where under certain conditions the model does not block transactions.
- D) Use statistical tests to check if there exists data drift.

Choose the right option: ____ C

11- One of the disadvantages of using A/B testing between the challenger model and champion model in production, is the difficulty of generating a data split without a sampling bias. (1 pts/ -0.25pts)

- A) True.
- B) False.

Choose the right option: ____ A